

Cloud Computing Applications and Services

(Aplicações e Serviços de Computação em Nuvem)

Guide 0: Warm-Up

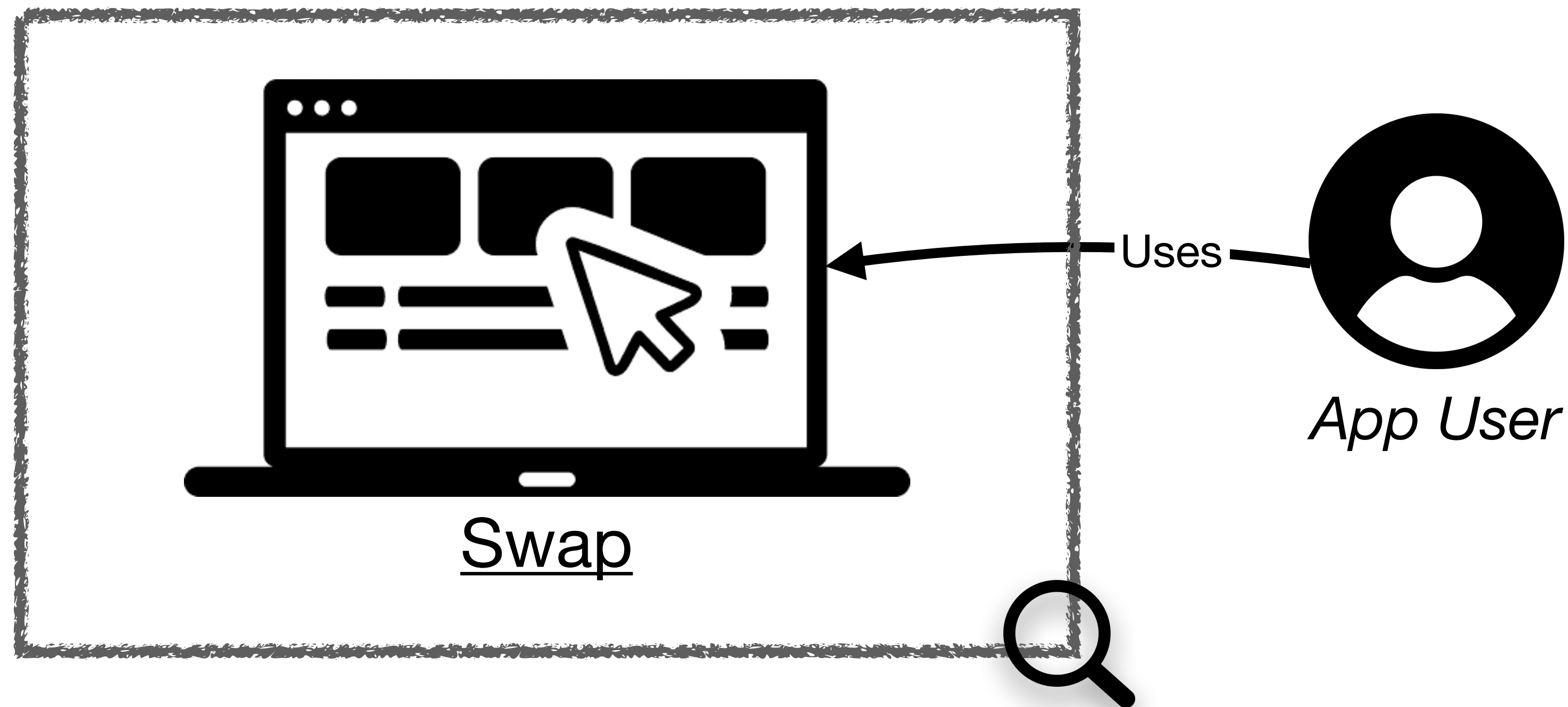
University of Minho

2026-2027

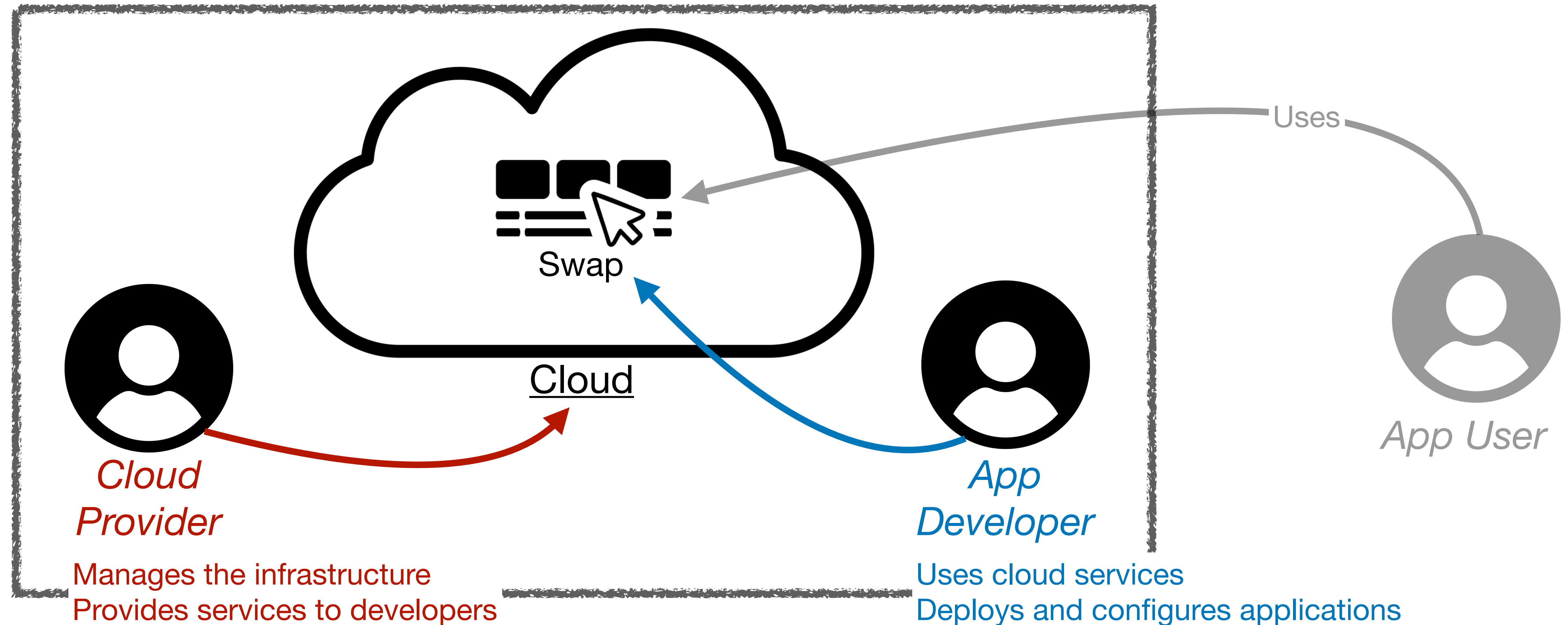


Context

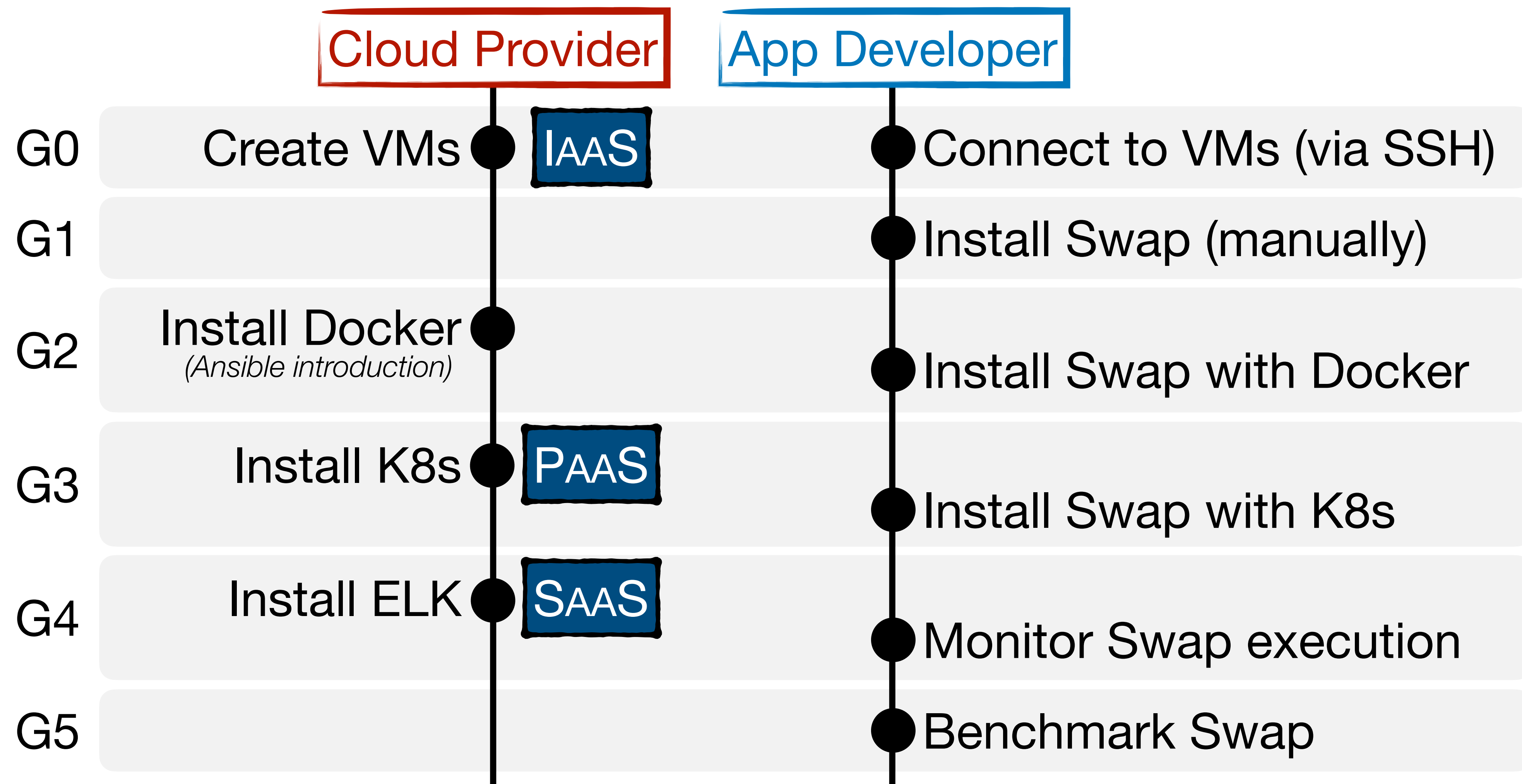
- The University of Minho hired you to deploy and manage Swap, a service for students swapping practical shifts.



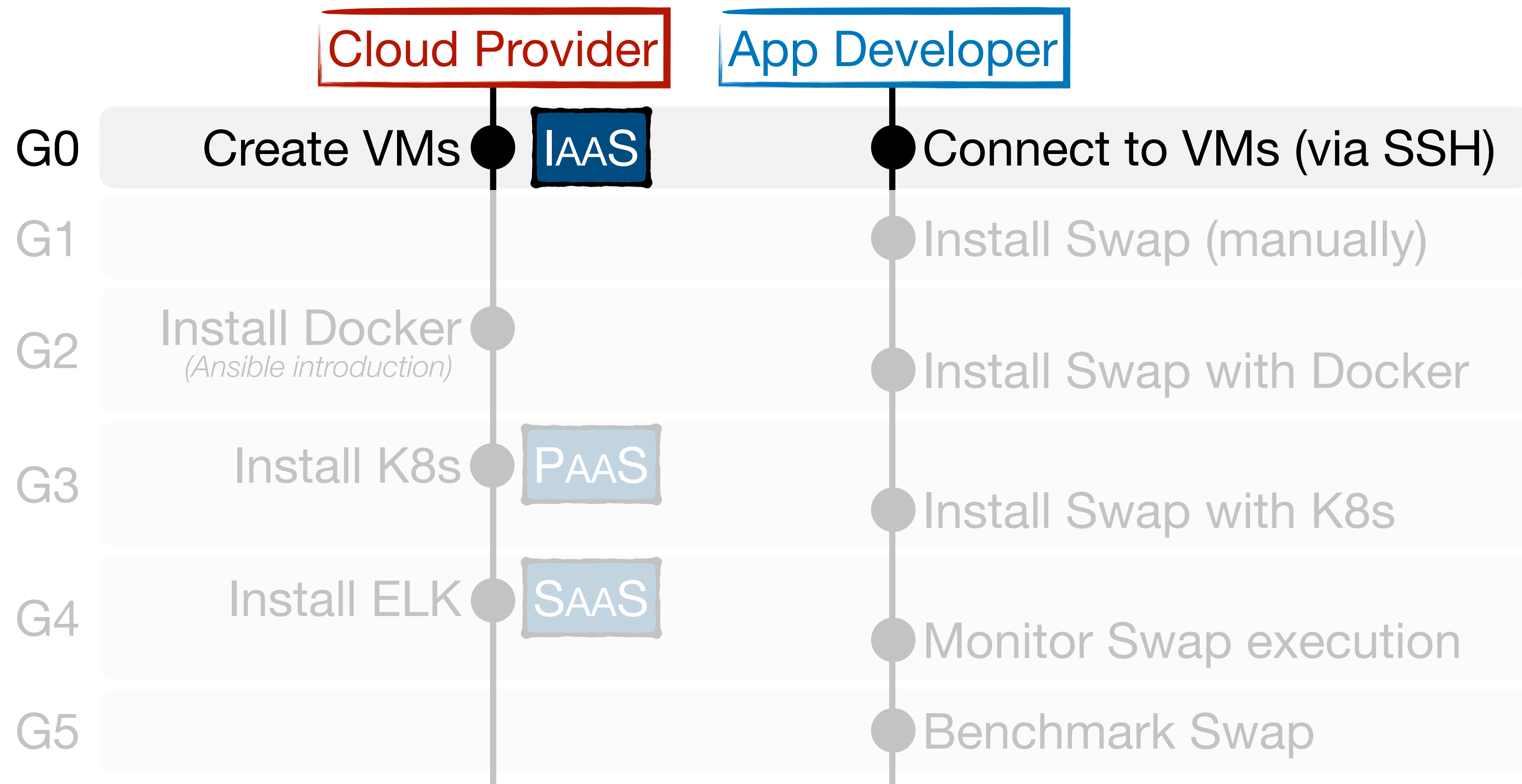
Context



Road Map



Road Map

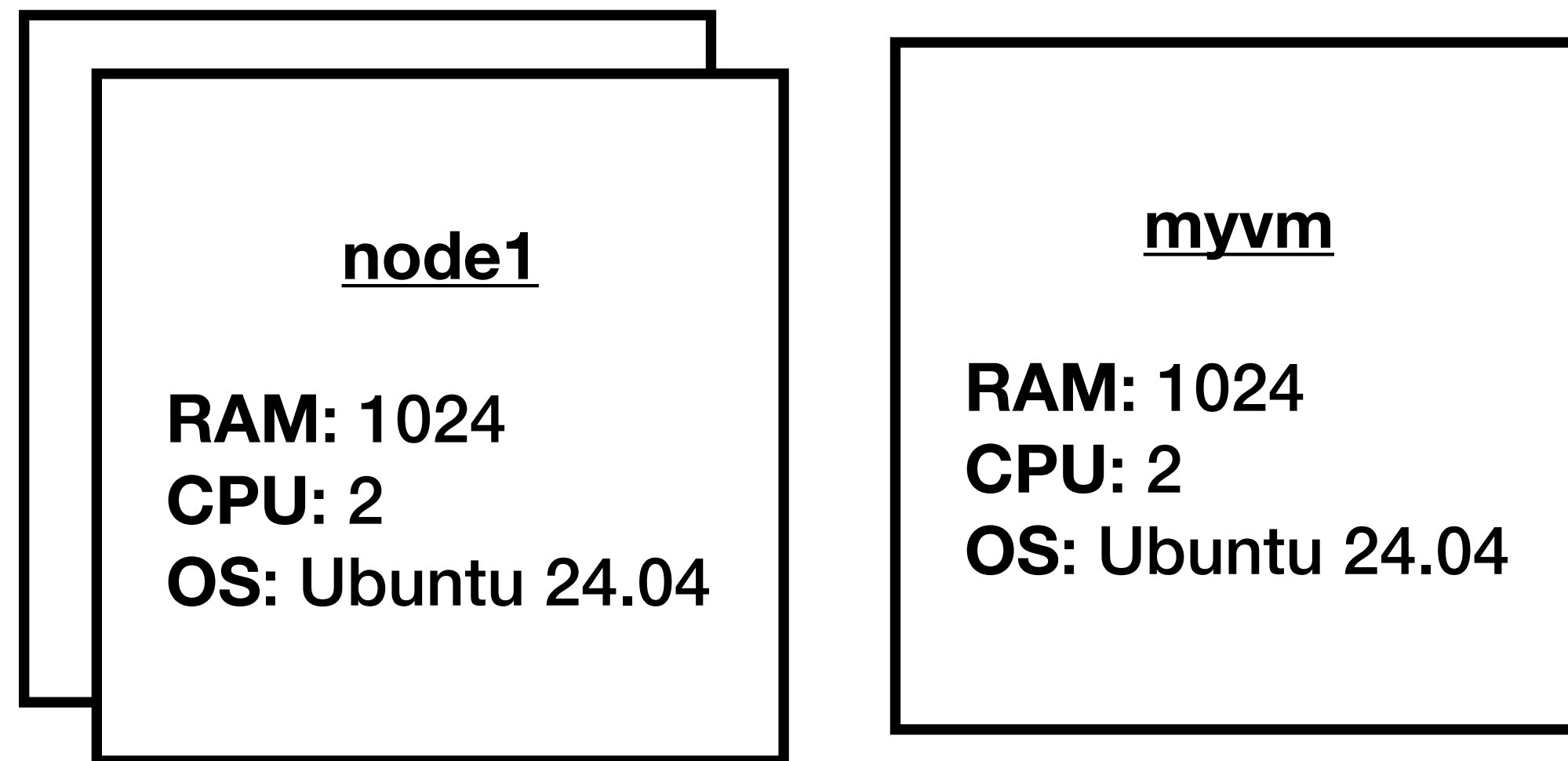


Goal

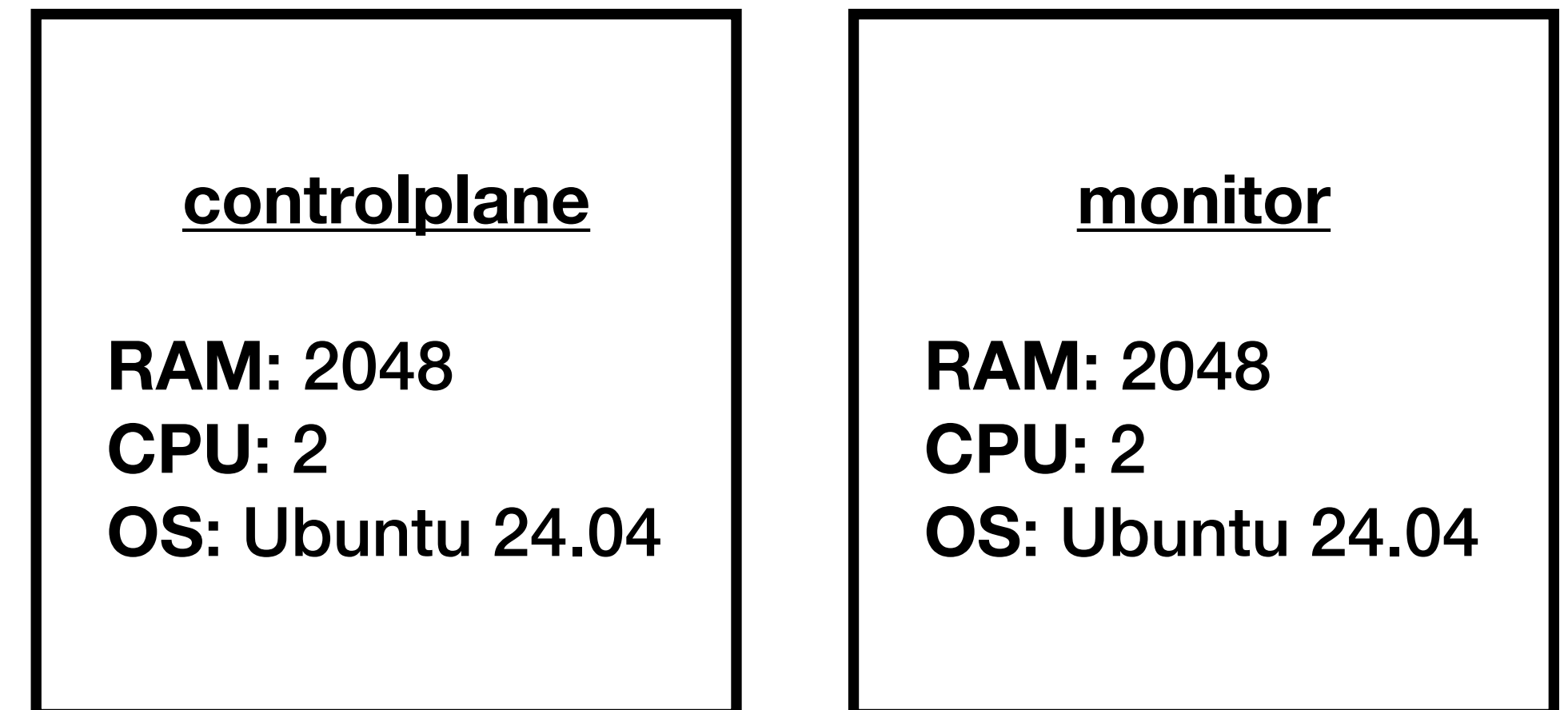
Cloud Provider

- Use Vagrant to create and configure 5 VirtualBox virtual machines.

Type 1



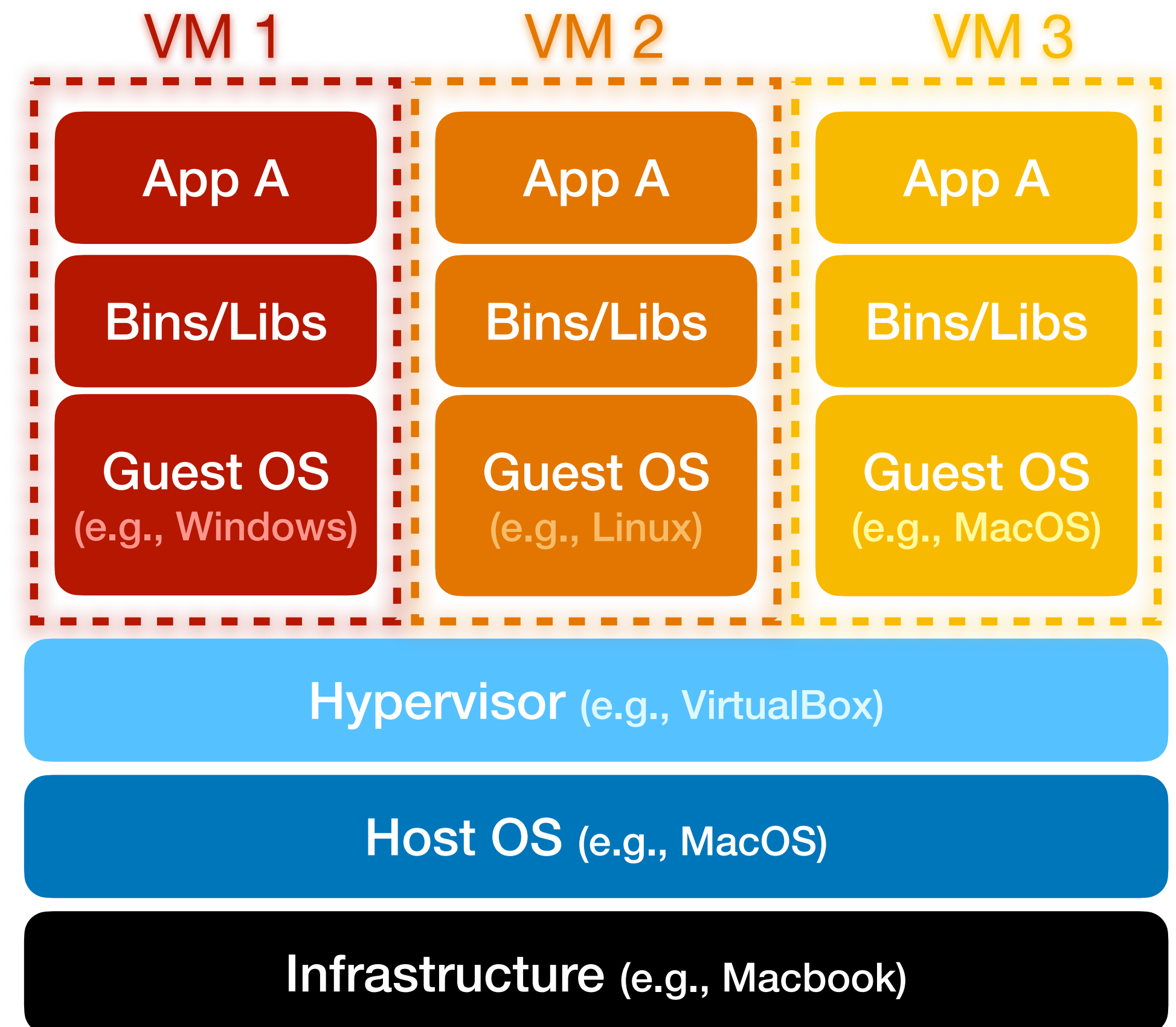
Type 2



Useful Tools and Concepts

VirtualBox

- VirtualBox is a virtualization software for running multiple operating systems in a single host.
- Want to know more about virtualization and virtual machines? Stay tuned for theoretical classes!



Vagrant

- Vagrant is a software for building and maintaining portable virtual software development environments (e.g., for VMware, VirtualBox, etc).
- Vagrant uses base images (known as boxes) to quickly clone a VM.
 - E.g., the box “bento/ubuntu-24.04” provides a base image of Ubuntu 24.04.
- The type of virtual machines and the tasks of how to configure and provision them are defined in a Vagrantfile.

Vagrant

Useful Commands

● Start the VMs:

- vagrant up
- vagrant up <vm_name>

● Stop the VMs:

- vagrant halt
- vagrant halt <vm_name>

● Output the status of the VMs:

- vagrant status
- vagrant status <vm_name>

● Provision the VMs (forces reprovisioning of the VMs):

- vagrant provision
- vagrant provision <vm_name>

● Clean up the VMs (stops and deletes the VMs):

- vagrant destroy
- vagrant destroy -f
- vagrant destroy <vm_name>
- vagrant destroy -f <vm_name>

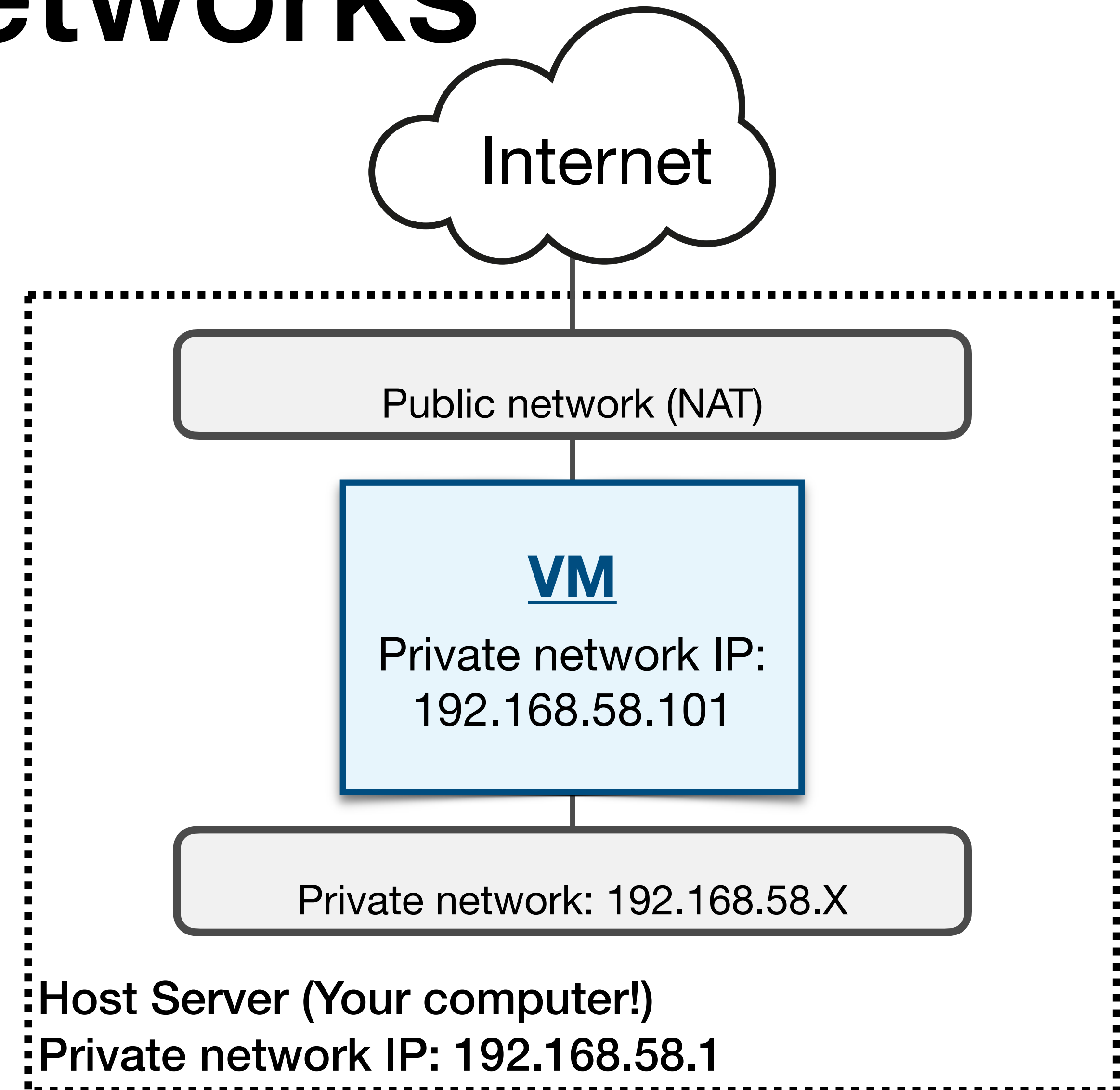
Virtual Networks

◎ NAT (Public network)

- VMs access the Internet through (masked as) the Host IP

◎ Host-only (Private Network)

- Only the Host and VMs can communicate with each other



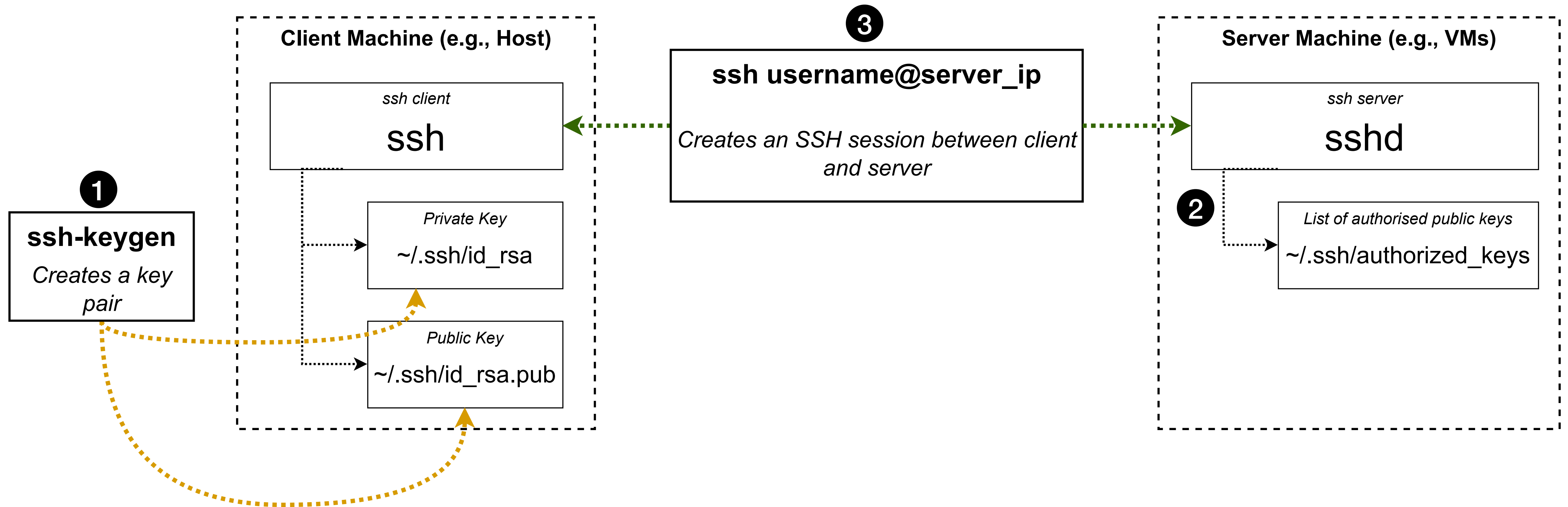
SSH

- **SSH (secure shell)** is a network protocol that allows secure remote access to systems over an unsecured network.
- SSH uses a client-server model, where the client connects to the server and establishes a secure channel for communication.
 - **SSH Server (sshd)**: program that runs on the remote system and listens for incoming SSH connections.
 - **SSH Client (ssh)**: program used to initiate an SSH connection to a remote server.

SSH

- **SSH keys:** SSH uses public-key cryptography to authenticate and establish secure connections.
 - The **private key (id_rsa/id_ed25519)** is secret, known only to the user, and should remain stored safely (e.g., on the user's laptop used to access the remote server).
 - The **public key (id_rsa.pub/id_ed25519.pub)** can be shared freely with any SSH server to which the user wishes to connect (e.g., VM).
 - The **authorized Keys** file (~/.ssh/authorized_keys on the server) specifies the SSH public keys that can be used to log in to the remote server.

SSH



Note: Step ❶ needs to be done manually. Step ❷ is specified at the Vagrantfile.

Example: For accessing *node1*, one should run `ssh vagrant@192.168.58.101` at the host's terminal (step ❸).

SSH

Useful Commands

● Generate new SSH keys:

- ssh-keygen

● Copy the Pub key to a remote server:

- ssh-copy-id <username>@<vm_ip>

● Connect to a remote server:

- ssh <username>@<vm_ip>

● Copy a file from local to remote:

- scp <local_file_path>
<username>@<vm_ip>:<remote_dst_path>

● Copy a folder from local to remote:

- scp -r <local_folder_path>
<username>@<vm_ip>:<remote_dst_path>

● Copy a file from remote to local:

- scp <username>@<vm_ip>:
<remote_file_path> <local_dst_path>

● Copy a folder from remote to local:

- scp -r
<username>@<vm_ip>:<remote_folder_path> <local_dst_path>